

U.G.C. NET Ex

COMPUTER

(Solved with Expl

Note: This paper contains *fifty* (50) objective type questions of two (2) marks each. All questions are compulsory. The candidates are required to select the most appropriate answer of each question.

1. COCOMO stands for

- (a) Composite Cost Model
- (b) Constructive Cost Model
- (c) Constructive Composite Model
- (d) Comprehensive Construction Model

Ans (b) The constructive cost model (COCOMO) is a procedural software cost estimation model developed by W. Boehm. The model uses a basic regression formula with parameters that are derived from historical project data and current as well as future project characteristics.

2. Match the following :

List-I

- (a) Good quality
- (b) Correctness
- (c) Predictable
- (d) Reliable

List-II

- 1. Program does not fail for a specified time in a given environment
- (2) Meets the functional requirements
- (3) Meets both functional and non-functional requirements
- (4) Process is under statistical control

Codes:

	A	B	C	D
(a)	3	2	4	1
(b)	2	3	4	1
(c)	1	2	4	3
(d)	1	2	3	4

Ans (a) Correctness meets the functional requirements. Good quality more than just correctness meets both functional and non-functions requirements.

Reliability is a characteristic which states that program does not fail for a specific time in given environment. Predictable means process in under statistical control and so is predictable.

3. While estimating the cost of software, Lines Of Code (LOC) and Function Points (FP) are used to measure which one of the following?

- (a) Length of code
- (b) Size of software
- (c) Functionality of software
- (d) None of the above

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Ans (b) Lines of Code (LOC) is a software metric used to measure the size of a computer program by counting the number of lines in the text of the program's source code.

A function point is a "unit of measurement" to express the amount of business functionally an information system (as a product) provides to a user. Function points are used to compute a functional size measurement of software. The cost (in dollars or hours) of a single unit is calculated from past projects.

4. A good software design must have

- (a) High module of coupling, High module cohesion
- (b) High module coupling, Low module cohesion
- (c) Low module coupling, High module cohesion
- (d) Low module coupling, Low module cohesion

Ans (c) Cohesion is a measure that defines the degree of intra-dependability within elements of a module.

Coupling is a measure that defines the lines of inter-dependability among modules of a program. It tells at what lines the modules interface and interact with each other.

All good software have low coupling and high cohesion.

5. Cyclometric complexity of a flow graph G with n vertices and e edges is

- (a) $V(G) = e + n - 2$
- (b) $V(G) = e - n + 2$
- (c) $V(G) = e + n + 2$
- (d) $V(G) = e - n - 2$

Ans (b) The cyclomatic complexity of a structured program is defined with reference to the control flow graph of the program, a directed graph containing the basic blocks of the program, with an edge between two basic blocks if control may pass from the first to the second. The complexity M is then defined as

$$M = E - N + 2P$$

where, E is the number of edges, N is the number of vertices and P is the number of connected components. For a single program (or subroutine or method), P is always equal to 1. So, a simpler formula for a single subroutine is:

$$M = E - N + 2$$

6. When the following code is executed what will be the value of x and y ?

int x = 1, y = 0

y = x++ ;

- (a) 2, 1
- (b) 2, 2
- (c) 1, 1
- (d) 1, 2

Ans (a) $y = x++$; returns y then increases x. Initially given x is 1. So, y is set to 1 and then x is incremented to 2.

7. How many values can be held by an array $A(-1, m; 1, m)$?

- (a) m (b) m^2
(c) $m(m+1)$ (d) $m(m+2)$

Ans (d) $A[-1, m] [1, m]$ is two dimensional array where,

Total number of rows = $m - (-1) + 1 = (m + 2)$

Total number of columns = $m - 1 + 1 = m$

Therefore, size of the array is = $m * (m + 2)$

8. What is the result of the expression $(1 \& 2) + 3/4$?

- (a) 1 (b) 2
(c) 3 (d) 4

Ans (d)

01	
AND	10
00	

$(1 \& 2)$ is a bitwise AND operation, i.e., $(3/4)$ give 4 as integer result.

Therefore, $(1 \& 2) + (3/4) = 0 + 4 = 4$

9. How many times the word 'print' shall be printed by the following program segment?

```
for (i = 1, i ≤ 2, i++)
for (j = 1, j ≤ 2, j++)
for (k = 1, k ≤ 2, k++)
Printf("print / n")
```

- (a) 1 (b) 3
(c) 6 (d) 8

Ans (d) For $i = 1, j = 1$
2 times the word will be printed.
for $i = 1, j = 2$
2 times the word will be printed.
for $i = 2, j = 1$
2 times the word will be printed.
for $i = 2, j = 2$
2 times the word will be printed.

10. Which of the following is not a type of Database Management System?

- (a) Hierarchical (b) Network
(c) Relational (d) Sequential

Ans (d) There are four structural types of database management systems.

1. Hierarchical database.
2. Network database.
3. Relational database.
4. Object oriented database.

But sequential is not a database model.

11. Manager's salary details are to be hidden from Employee Table. This Technique is called as

- (a) Conceptual level Datahiding
(b) Physical level Datahiding
(c) External level Datahiding
(d) Logical level Datahiding

Ans (c) **Data hiding** is a software development technique used in OOP. It is mainly used to hide internal object details, i.e., the design decisions in a program that are most likely to change.

There are three form of schema, physical conceptual and external.

External data hiding is tailored to the needs of a particular category of users. Portions of stored data should not be seen by some users and begins to implement a level of security and simplifies the view for these users.

12. A Network Schema

- (a) restricts to one to many relationship
- (b) permits many to many relationship
- (c) stores Data in a Database
- (d) stores Data in a Relation

Ans (b) 1. Network model permits the modeling of many to many relationships in data. A set consists of an owner record type, a set name, and member a member record type.

2. The network data structure looks like a tree structure, except that a dependent node called a child node may have more than one parent or owner node. So, one or more nodes may have multi-parents. Therefore a network model allows a more natural modeling of relationship between entities. There is no superior or subordinate relationship in network model as exists in hierarchical models.

13. Which normal form is considered as adequate for usual database design?

- (a) 2 NF
- (b) 3 NF
- (c) 4 NF
- (d) 5 NF

Ans (b) BCNF does not guarantee dependency preserving, but 3 NF satisfied both dependency preserving and lossless join. These are goal of database design.

So, 3 NF is considered as adequate normal form.

14. If $D_1, D_2 \dots D_n$ are domains in a relational model, then the relation is a table, which is a subset of

- (a) $D_1 + D_2 + \dots + D_n$
- (b) $D_1 \times D_2 \times \dots \times D_n$
- (c) $D_1 \cup D_2 \cup \dots \cup D_n$
- (d) $D_1 - D_2 - \dots - D_n$

Ans (b) The (x) Cartesian product operator is a binary operator to combines the tuples of the relation with all the tuples of the other relation.

The value of relation is a subset of the cartesian product (x) of $D_1, D_2, \dots, D_n$.

15. Which of the following addresses is used to deliver a message to the correct application program running on a host?

- (a) Port
- (b) IP
- (c) Logical
- (d) Physical

Ans (a) A port number is a way to identity a specific process to which an Internet or other network message is to be forwarded when it arrives at a server. For the TCP (transmission control protocol) and UDP (user datagram protocol), a port number is a 16 bit integer that is put in the header appended to a message unit.

Specific port numbers are often used to identity specific services.

16. In _____ substitution, a character in the plaintext is always changed to the same character in the ciphertext, regardless of its position in the text.

- (a) polyalphabetic (b) monoalphabetic
(c) transpositional (d) multialphabetic

Ans (b) In monoalphabetic substitution, a character (or a symbol) in the plain text is always changed to the same character (or symbol) in ciphertext regardless of its position in the text. In other words, the relationship between letters in the plaintext and the ciphertext is one to one.

Whereas polyalphabetic is one to many.

17. In classful addressing, the IP address 190.255.254.254 belongs to

- (a) Class A (b) Class B
(c) Class C (d) Class D

Ans (b) Range of IPv4 addresses:

Class	Range (First octet in Decimal)	First octet (from MSB)
A	1-126	0
B	128-191	10
C	192-223	110
D	224-239	1110
E	240-255	1111

So, first octet (190) of given IP address (190.225.254.254) belongs to class B.

18. In hierarchical routing with 4800 routers, what region and cluster size should be chosen to minimize the size of the routing table for a three layer hierarchy?

- (a) 10 clusters, 24 regions and 20 routers
(b) 12 clusters, 20 regions and 20 routers
(c) 16 clusters, 12 regions and 25 routers
(d) 15 clusters, 16 regions and 20 routers

Ans (d) Assume that there is one 'a' clusters, 'b' regions in each region and 'c' routers per region.

And $abc = 4800$ (all the options quality multiplications rule).

Now, the formula to minimize is

$$(a - 1) + (b - 1) + c$$

Therefore,

- (a) sums 52
(b) sums 50
(c) sums 51
(d) sums 49

So, in option (d) will have to keep minimum routing table.

19. In IPv4 header, the _____ field is needed to allow the destination host to determine which datagram a newly arrived fragments belongs to.

- (a) identification
(b) fragment offset
(c) time to live
(d) header checksum

Ans (a) 1. The **identification field** is needed to allow the destination host to determine which datagram a newly arrived fragment belongs to all the fragments of a datagram contain the same identification value.

2. **Time of live (TTL)** field is a counter used to limit packet lifetimes.

3. The **header checksum** verifies the header only.

20. Given $L_1 = L(a^*baa^*)$ and $L_2 = L(ab^*)$. The regular expression corresponding to language $L_3 = L_1/L_2$ (right quotient) is given by

- (a) a^*b (b) a^*baa^*
(c) a^*ba^* (d) None of these

Ans (c) The right quotient (or simply) quotient of a formal language L_1 with a formal language L_2 is the language consisting of strings w such that wx is in L_1 for some string x in L_2 i.e.,

$$L_1/L_2 = \{w | \exists x((x \in L_2) \wedge (wx \in L_1))\}$$

Since given

$$L_1 = a^*baa^* = \{ba, aba, abaa, baa, \dots\}$$

$$L_2 = ab^* = \{a, ab, abb, abbb, \dots\}$$

Therefore,

$$L_1/L_2 = \{b, ab, ba, aba, \dots\} = (a^*ba^*)$$

we have used only string 'a' of language L_2 .

21. Given the production rules of a grammar G_1 as

$$S_1 \rightarrow AB \mid aaB$$

$$A \rightarrow a \mid Aa$$

$$B \rightarrow b$$

and the production rules of a grammar G_2 as

$$S_2 \rightarrow aS_2bS_2 \mid bS_2aS_2 \mid \lambda$$

Which of the following is correct statement?

- (a) G_1 is ambiguous and G_2 is not ambiguous.
(b) G_1 is ambiguous and G_2 ambiguous.
(c) G_1 is not ambiguous and G_2 is ambiguous.
(d) G_1 is not ambiguous and G_2 is not ambiguous.

Ans (b) An **ambiguous grammar** is a context free grammar for which there exists a string that can have more than one parse tree.

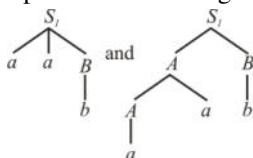
Given, grammar G_1 is

$$S_1 \rightarrow AB \mid aaB$$

$$A \rightarrow a \mid Aa$$

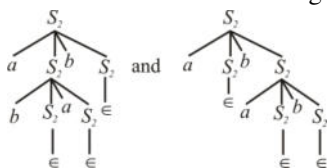
$$B \rightarrow b$$

G_1 can have two parse tree for string $w = aab$.



Grammar G_2 is $S_2 \rightarrow aS_2bS_2 \mid bS_2aS_2 \mid \epsilon$

G_2 can have two derivation tree for string $w = abab$.



Both grammar G_1 and G_2 are ambiguous grammar.

22. Given a grammar: $S_1 \rightarrow Sc, S \rightarrow SA \mid A, A \rightarrow aSb \mid ab$, there is a rightmost derivation $S_1 \Rightarrow Sc \Rightarrow SAC \Rightarrow SaSbc$.

Thus, $SaSbc$ is a right sentential form, and its handle is

- (a) SaS (b) bc
(c) Sbc (d) aSb

Ans (d)

$SaSbc$

$Sac [A \rightarrow aSb]$

$Sc[S \rightarrow SA]$

$S_1[S_1 \rightarrow Sc] \Rightarrow aSb$

23. The equivalent production rules corresponding to the production rules $S \rightarrow S\alpha_1 \mid S\alpha_2 \mid \beta_1 \mid \beta_2$ is

- (a) $S \rightarrow \beta_1 \mid \beta_2, A \rightarrow \alpha_1 A \mid \alpha_2 A \mid \lambda$
(b) $S \rightarrow \beta_1 \mid \beta_2 \mid \beta_1 A \mid \beta_2 A, A \rightarrow \alpha_1 A \mid \alpha_2 A$
(c) $S \rightarrow \beta_1 \mid \beta_2, A \rightarrow \alpha_1 A \mid \alpha_2 A$
(d) $S \rightarrow \beta_1 \mid \beta_2 \mid \beta_1 A \mid \beta_2 A, A \rightarrow \alpha_1 A \mid \alpha_2 A \mid \lambda$

Ans (d) $S \rightarrow S\alpha_1 \mid S\alpha_2 \mid \beta_1 \mid \beta_2$

Left factoring the given grammar gives.

$S \rightarrow \beta_1 \mid \beta_2 \mid A\beta_1 \mid A\beta_2.$

$A \rightarrow \alpha_1 A \mid \alpha_2 A \mid \lambda$

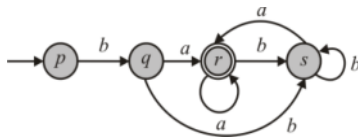
24. Given a Non-deterministic Finite Automation (NFA) with states p and r as initial and final states respectively and transition table as given below:

	a	b
p	—	q
q	r	s
r	r	s
s	r	s

The minimum number of states required in Deterministic Finite Automation (DFA) equivalent to NFA is

- (a) 5 (b) 4
(c) 3 (d) 2

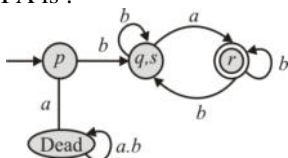
Ans (b) NFA will be



Transition table of DFA will be

	a	b
$\rightarrow p$	Dead	q
q	r	s
r	r	s
s	r	s
Dead	Dead	Dead

We can merge state q and s as same transition.
So, required DFA is :



25. Which is the correct statement(s) for Non Recursive predictive parser?

- S_1 : **First** (α) = $\{t \mid \alpha \Rightarrow t\beta \text{ for some string } \beta\} \Rightarrow t\beta$
 S_2 : **Follow** (X) = $\{a \mid S \Rightarrow aXa\beta \text{ for some string } \alpha \text{ and } \beta\}$
 (a) Both statements S_1 and S_2 are incorrect.
 (b) S_1 is incorrect and S_2 is correct.
 (c) S_1 is correct and S_2 is incorrect.
 (d) Both statements S_1 and S_2 are correct.

Ans (d) We associate each grammar symbol A with the first (A). The implication of this set is that the grammar symbol A can in some steps of transition produce the elements of the set first (A). Follow is defined only for non-terminals of the grammar G. It can be defined as the set of terminals of grammar G. which can immediately follow the non-terminal in a production rule from start symbol. Both statements are true.

26. Given an open address hash table with load factor $\alpha < 1$, the expected number of probes in a successful search is

- (a) Atmost $\frac{1}{\alpha} \ln \left(\frac{1-\alpha}{1} \right)$
 (b) Atmost $\frac{1}{\alpha} \ln \left(\frac{1}{1-\alpha} \right)$
 (c) Atmost $\frac{1}{\alpha} \ln \left(\frac{1}{1-\alpha} \right)$
 (d) Atmost $\frac{1}{\alpha} \ln \left(\frac{\alpha}{1-\alpha} \right)$

Ans (b) Theorem: Given an open address hash table with load factor $\alpha = n/m < 1$, the expected number of probes in an unsuccessful and a successful search are at most $1/(1-\alpha)$ and $(1/\alpha) \ln (1/(1-\alpha))$ respectively. Assuming uniform hashing and assuming that each key in the table is equally likely to be searched for.

27. For a B-Tree of height h and degree t , the total CPU time used to insert a node is

- (a) $O(h \log t)$ (b) $O(t \log h)$
 (c) $O(t^2 h)$ (d) $O(th)$

Ans (d) Inserting a key K into a B-tree T of height h is done in a single pass down tree, requiring $O(h)$ disk accesses. The CPU time required is $O(t.h) = O(t \cdot \log_t(n))$.

28. The time complexity to build a heap with a list of n -numbers is

- (a) $O(\log n)$ (b) $O(n)$
 (c) $O(n \log n)$ (d) $O(n^2)$

Ans (b) The time complexity of build heap tree procedure is $O(n)$.

29. The value of postfix expression:

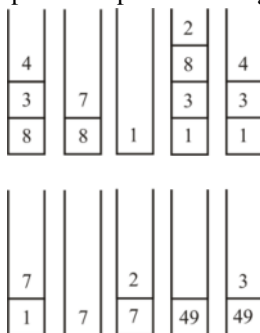
$834 + - 382 / + \times 2 \$ 3 +$ is

- (a) 17 (b) 131
(c) 64 (d) 52

Ans (d) Given, postfix expression is

$834 + - 382 / + * 2 \$ 3 +$

Evaluation of postfix expression using operand stack is



30. Consider the following statements for priority queue :

S_1 : It is data structure in which the intrinsic ordering of the elements does determine the result of the elements does determine the result of its basic operations.

S_2 : The elements of a priority queue may be complex structures that are ordered on one or several fields.

Which of the following is correct?

- (a) Both S_1 and S_2 are correct.
(b) S_1 is correct and S_2 is incorrect.
(c) S_1 is incorrect and S_2 is correct.
(d) Both S_1 and S_2 are correct.

Ans (d) 1. A **priority queue** is a data structure in which the intrinsic ordering of the elements determines the results of its basic operations.

2. There are two types of priority queues:

ascending order and **descending order** priority queues. The elements of a priority queue need not be numbers or characters that can be compared directly. They may be complex structures that are ordered on one or several fields.

31. Repository of information gathered from multiple sources, storing under unified scheme at a single site is called as.

- (a) Data mining (b) Meta data
(c) Data warehousing (d) Database

Ans (c) 1. A **data warehouse** is constructed by integrating data from multiple heterogeneous sources. It supports analytical reporting, structured and/or adhoc queries and decision making.

2. Collections of database that work together are called data warehouses. This makes in possible to integrate data from multiple database.

3. Data mining is used to help individuals and organizations make better decisions. Meta data is data that describes other data.

32. The task of correcting and pre processing data is called as

- (a) Data streaming (b) Data cleaning
- (c) Data mining (d) Data storming

Ans (b) Data scrubbing, also called data cleansing is the process of amending or removing data in a database that is incorrect, incomplete, improperly formatted, or duplicated.

33. Using data $p = 3, q = 11, n = pq, d = 7$ in RSA algorithm find the cipher text of the given plain text SUZANNE:

- (a) BUTAE EZ (b) SUZANNE
- (c) XYZABCD (d) ABCDXYZ

Ans (a) Given, $P = 3, q = 11, d = 7, n = 33$.
 Therefore, using RSA algorithm

$$\phi(n) = (P-1)(q-1) = 2 \times 10 = 20$$
 and $ed = 1 \bmod \phi(n)$
 $G(1)(e, \phi(n)) = 1$
 so, $e = 3, d = 7$
 Now, $a = 1, b = 2, c = 3, \dots, z = 26$
 Given SUZANNE = 19, 21, 26, 1, 14, 14, 5
 Since, $m' = m^e \bmod N$.
 So,
 $\Rightarrow 19^3 \bmod 33 = 28$ (i.e., $28 - 26 = 2$)
 $= 2 \Rightarrow B$
 $\Rightarrow 21^3 \bmod 33 = 21 \Rightarrow U$
 $\Rightarrow 26^3 \bmod 33 = 20 \Rightarrow T$
 $\Rightarrow 1^3 \bmod 33 = 1 \Rightarrow A$
 $\Rightarrow 14^3 \bmod 33 = 5 \Rightarrow E$
 $\Rightarrow 5^3 \bmod 33 = 26 \Rightarrow Z$
 Therefore ciphertext is BUTAE EZ.

34. The relation "divides" on a set of positive integers is _____.

- (a) Symmetric and transitive
- (b) Anti symmetric and transitive
- (c) Symmetric only
- (d) Transitive only

Ans (b) 1. A relation R on a set A is called antisymmetric if and only if for any a and b in A , whenever $(a, b) \in R$ and $(b, a) \in R$ then $a = b$ must hold.
2. Since, a/b and b/a is true if and only if $a = b$ then the relation satisfies the property of being antisymmetric.
3. A relation R on a set A is called transitive if $(a, b) \in R$ and $(b, c) \in R$ the $(a, c) \in R$ must hold.
4. If a/b and b/c hold then a/c also hold so, given relation is antisymmetric and transitive.

35. Give as good a big- O estimate as possible for the following functions:

- $(n \log n + n^2) (n^3 + 2)$ and $(n! + 2^n) (n^3 + \log (n^2 + 1))$
- (a) $O(n^5 + 2n^2)$ and $O(n^3 \times n!)$
- (b) $O(n^5)$ and $O(n^3 \times 2^n)$

- (c) $O(n^5)$ and $O(n^3 \times n!)$
 (d) $O(n^5 + 2n^2)$ and $O(n^3 \times 2^n)$

Ans (c) Let f and g be two functions defined on some subset of the real numbers one writes $f(x) = O(g(x))$ as $x \rightarrow 0$. If and only if there is a positive constant m such that for all sufficiently large value of x . In other words
 $\Rightarrow |f(x)| \leq m|g(x)| \forall x \geq x_0$
 $\Rightarrow f(x) = O(g(x))$
 $\Rightarrow \lim_{n \rightarrow \infty} \frac{f(x)}{g(n)} = 0$

36. A test contains 100 true/false questions. How many different ways can a student answer the questions on the test, if the answer may be left blank also.

- (a) $^{100}P_2$ (b) $^{100}C_2$
 (c) 2^{100} (d) 3^{100}

Ans (d) There is three options of each question, i.e., either answer true or answer false or leave it blank. So, total number of answering the test.
 $= 3 \times 3 \times 3 \times \dots \times (100 \text{ times}) = 3^{100}$

37. Which of the following connected simple graph has exactly one spanning tree?

- (a) Complete graph (b) Lord Harvey
 (c) Thomas Shadwell (d) Joseph Addison

Ans (d) A connected simple graph has unique spanning tree only if graph edges have distinct edge weights.

38. How many edges must be removed to produce the spanning forest of a graph with N vertices, M edges and C connected components?

- (a) $M + N - C$ (b) $M - N - C$
 (c) $M - N + C$ (d) $M + N + C$

Ans : (c) $[M - N + C]$ edges must be remove to produce to spanning forest of a graph with 'N' vertices.

39. Which of the following shall be a compound proposition involving the propositions p, q and r, that is true when exactly two of the p, q and r are true and is false otherwise?

- (a) $(p \vee q \wedge \neg r) \vee (p \wedge q \vee r) \wedge (\neg p \wedge q \vee r)$
 (b) $(p \wedge q \vee r) \wedge (p \wedge q \wedge r) \vee (\neg q \wedge \neg p \wedge \neg r)$
 (c) $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$
 (d) $(p \vee r \wedge q) \vee (p \wedge q \wedge r) \vee (\neg p \wedge q \wedge r)$

Ans : (c) Draw the truth table of boolean function that involves p, q and r that is true when exactly two of the p, q and r are true and is false otherwise.

p	q	r	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Therefore,

$$f = (\neg p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge q \wedge \neg r)$$

40. The truth value of the statements:

$\exists!xP(x) \rightarrow \exists xP(x)$ and $\exists!x\neg P(x) \rightarrow \neg \forall xP(x)$

(Where the notation $\exists!xP(x)$ denotes the proposition "There exists a unique x such that $P(x)$ is true") are:

- (a) True and False (b) False and True
(c) False and False (d) True and True

Ans : (d) (i) If there exists a unique x with $p(x)$ true, then there exist x with $p(x)$ true. This is true as exactly one is a subset of at least one.

(ii) If there exists a unique x with $p(x)$ false, then there exist an x with $p(x)$ false. This is also true since exactly one is a subset of atleast one and $\neg \forall x p(x) = \exists x \neg p(x)$.

41. How many different Boolean functions of degree 4 are there?

- (a) 2^4 (b) 2^8
(c) 2^{12} (d) 2^{16}

Ans : (d) Total number of boolean functions with total 2^n

Combinations = 2^{2^n} .

Given, $n = 4$, therefore,

Number of boolean functions = $2^{2^4} = 2^{16}$.

42. A Boolean operator θ is defined as follows:

$1\theta 1=1, 1\theta 0=0, 0\theta 1=0$ and $0\theta 0=1$

What will be the truth value of the expression

$(x\theta y)\theta z = x\theta(y\theta z)$?

- (a) Always false
(b) Always true
(c) Sometimes true
(d) True when x, y, z are all true

Ans : (b) $\overline{(xy + \bar{x}\bar{y})} = x\theta y$ always true.

43. Which one of the following is decimal value of a signed binary number 1101010, if it is in 2's complement form?

- (a) -42 (b) -22
(c) -21 (d) -106

Ans : (b) If MSB (most significant bit) of 2's complement representation is 1 then number is negative (magnitude is 2's complement) else number is positive itself.

Since MSB of given 2's complement representation is 1, so, given number is negative and its magnitude is:

$\Rightarrow$ 2's complement of 1101010

$\Rightarrow 0010101 + 1 = (0010110)_2$

$\Rightarrow (22)_{10}$

So, number is - 22.

44. A set of processors $P_1, P_2, \dots, P_k$ can execute in parallel if Bernstein's conditions are satisfied on a pairwise basis; that is $P_1 \parallel P_2 \parallel P_3 \dots \parallel P_k$ if and only if:

- (a) $P_i \parallel P_j$ for all $i \neq j$
- (b) $P_i \parallel P_j$ for all $i = j + 1$
- (c) $P_i \parallel P_j$ for all $i \leq j$
- (d) $P_i \parallel P_j$ for all $i \geq j$

Ans : (a) A set of processes can execute in parallel if Bernstein's conditions are satisfied on a pair wise basis; that is $P_1 \parallel P_2 \dots P_k$ if and only if $P_i \parallel P_j$ for all $i \neq j$.

- 45. When a mobile telephone physically moves from one of the another cell, the base station transfers ownership to the cell getting strongest signal. This process is known as _____ .**
- (a) handoff
 - (b) mobile switching
 - (c) mobile routing
 - (d) cell switching

Ans : (a) In a cellular telephone network, handoff is the transition for any given user of signal transmission from one base station to a geographically adjacent base station as the user moves around.

- 46. A virtual memory based memory management algorithm partially swaps out a process. This is an example of**
- (a) short term scheduling
 - (b) long term scheduling
 - (c) medium term scheduling
 - (d) mutual exclusion

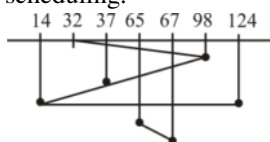
Ans : (c) Medium term scheduling is a part of swapping. It removes the processes from the memory. It reduces the degree of multiprogramming. The medium term scheduler is in charge of handling the swapped-out processes.

A long term schedules determines which program are admitted to the system for processing. It select processes from the queue and load them into memory for execution.

The short-term scheduler decides which of the ready, in-memory processes is to be executed after a clock interrupt, an I/O interrupt, an operating system call, or another form of signal.

- 47. Assuming that the disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O block requests are 98, 37, 14, 124, 65, 67:**
- (a) 310
 - (b) 324
 - (c) 320
 - (d) 321

Ans : (d) Given request is: 98, 37, 14, 124, 65, 67
Using FCFS (first come first serve) disk scheduling.



So, total number of disk moves:
= $66 + 84 + 110 + 59 + 2 = 321$

(a) 21 ns (b) 23 ns
(c) 30 ns (d) 35 ns

49. Consider the following UNIX command:

```
sort <in> temp; head - 30 <temp; rm temp
```

Which of the following functions shall be performed by this command?

- (a) Sort, taking the input from "temp", prints 30 lines from temp and delete the file temp
- (b) Sort the file "temp", removes 30 lines from temp and delete the file temp
- (c) Sort, taking the input from "in" and writing the output to "temp" then prints 30 lines from temp on terminal. Finally "temp" is removed.
- (d) Sort, taking the input from "temp" and then prints 30 lines from "temp" on terminal. Finally "temp" is removed.

50. The mv command changes

- (a) the inode
- (b) the inode-number
- (c) the directory entry
- (d) both the directory entry and the inode

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